

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <termios.h>
#include <unistd.h>

#define MAX_HISTORY 10
#define BUFFER_SIZE 100

char *history[MAX_HISTORY];
int history_count = 0;

void add_history(const char *command)
{
    if (strlen(command) == 0)
        return;

    if (history_count < MAX_HISTORY)
    {
        history[history_count] = strdup(command);
        history_count++;
    }
    else
    {
        free(history[0]);

        for (int i = 1; i < MAX_HISTORY; i++)
            history[i - 1] = history[i];

        history[MAX_HISTORY - 1] = strdup(command);
    }
}

void show_history()
{
    printf("\n\nCommand History:\n");

    for (int i = 0; i < history_count; i++)
        printf("%d: %s\n", i + 1, history[i]);
}
```

```
void free_history()
{
    for (int i = 0; i < history_count; i++)
        free(history[i]);
}

int main()
{
    char buffer[BUFFER_SIZE];

    printf("Simple Command History\n");
    printf("Type commands and press Enter.\n");
    printf("Type 'history' to display history.\n");
    printf("Type 'exit' to quit.\n\n");

    while (1)
    {
        printf("shell> ");
        fflush(stdout);

        if (fgets(buffer, BUFFER_SIZE, stdin) == NULL)
            break;

        buffer[strcspn(buffer, "\n")] = '\0';

        if (strcmp(buffer, "exit") == 0)
            break;

        if (strcmp(buffer, "history") == 0)
        {
            show_history();
            continue;
        }

        add_history(buffer);

        printf("Command entered: %s\n", buffer);
    }

    free_history();
}
```

```
    return 0;  
}
```

```
shell> ls  
Command entered: ls  
shell> pwd  
Command entered: pwd  
shell> hello  
Command entered: hello  
shell> test  
Command entered: test  
shell> history
```

Command History:

```
1: ls  
2: pwd  
3: hello  
4: test  
shell> |
```

GNU nano 7.2

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define INITIAL_SIZE 10

int main()
{
    char *buffer;
    int size = INITIAL_SIZE;

    // 1. Allocate buffer dynamically
    buffer = (char *)malloc(size * sizeof(char));

    if (buffer == NULL)
    {
        printf("Memory allocation failed!\n");
        return 1;
    }

    printf("Enter a string: ");
    fgets(buffer, size, stdin);

    // Remove newline
    buffer[strcspn(buffer, "\n")] = '\0';

    printf("You entered: %s\n", buffer);

    // 2. Resize buffer
    size = 20;

    char *temp = realloc(buffer, size * sizeof(char));

    if (temp == NULL)
```

```

    {
        free(buffer);
        printf("Memory reallocation failed!\n");
        return 1;
    }

    buffer = temp;

    printf("Enter another string: ");
    fgets(buffer, size, stdin);

    buffer[strcspn(buffer, "\n")] = '\0';

    printf("New string: %s\n", buffer);

    // 3. Release memory
    free(buffer);
    buffer = NULL;

    printf("Memory released successfully.\n");

    return 0;
}

```

```

sravya@Sravya1955:~/command_history$ nano memory_management.c
sravya@Sravya1955:~/command_history$ gcc -Wall -Wextra -g memory_management.c -o memory_management
sravya@Sravya1955:~/command_history$ ./memory_management
Enter a string: hello
You entered: hello
Enter another string: this is a longer string
New string: this is a longer st
Memory released successfully.
sravya@Sravya1955:~/command_history$ |

```

```
sravya@Sravya1955:~/command_history$ nano memory_management.c
sravya@Sravya1955:~/command_history$ N
N: command not found
sravya@Sravya1955:~/command_history$ rm memory_management.c
sravya@Sravya1955:~/command_history$ nano memory_management.c
sravya@Sravya1955:~/command_history$ ~/command_history$
-bash: /home/sravya/command_history$: No such file or directory
sravya@Sravya1955:~/command_history$ gcc -Wall -Wextra -g memory_management.c -o memory_management
sravya@Sravya1955:~/command_history$ ./memory_management
Dynamic Memory Management
-----
Enter strings. Type 'exit' to finish.

Enter string: hello

Enter string: this is a longer string

Enter string: another test string

Enter string: exit

Dynamic Array:
1: hello
2: this is a longer string
3: another test string

Linked List:
hello
this is a longer string
another test string

All memory released successfully.
sravya@Sravya1955:~/command_history$ |
```

```
--797--
Dynamic Memory Management
-----
Enter strings. Type 'exit' to finish.

Enter string: hello

Enter string: this is a longer string

Enter string: another test string

Enter string: exit

Dynamic Array:
1: hello
2: this is a longer string
3: another test string

Linked List:
hello
this is a longer string
another test string

All memory released successfully.
==797==
==797== HEAP SUMMARY:
==797==      in use at exit: 0 bytes in 0 blocks
==797==    total heap usage: 20 allocs, 20 frees, 2,364 bytes allocated
==797==
==797== All heap blocks were freed -- no leaks are possible
==797==
==797== For lists of detected and suppressed errors, rerun with: -s
==797== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
sravya@Sravya1955:~/command_history$ |
```